

$-2 \Delta \ln L$

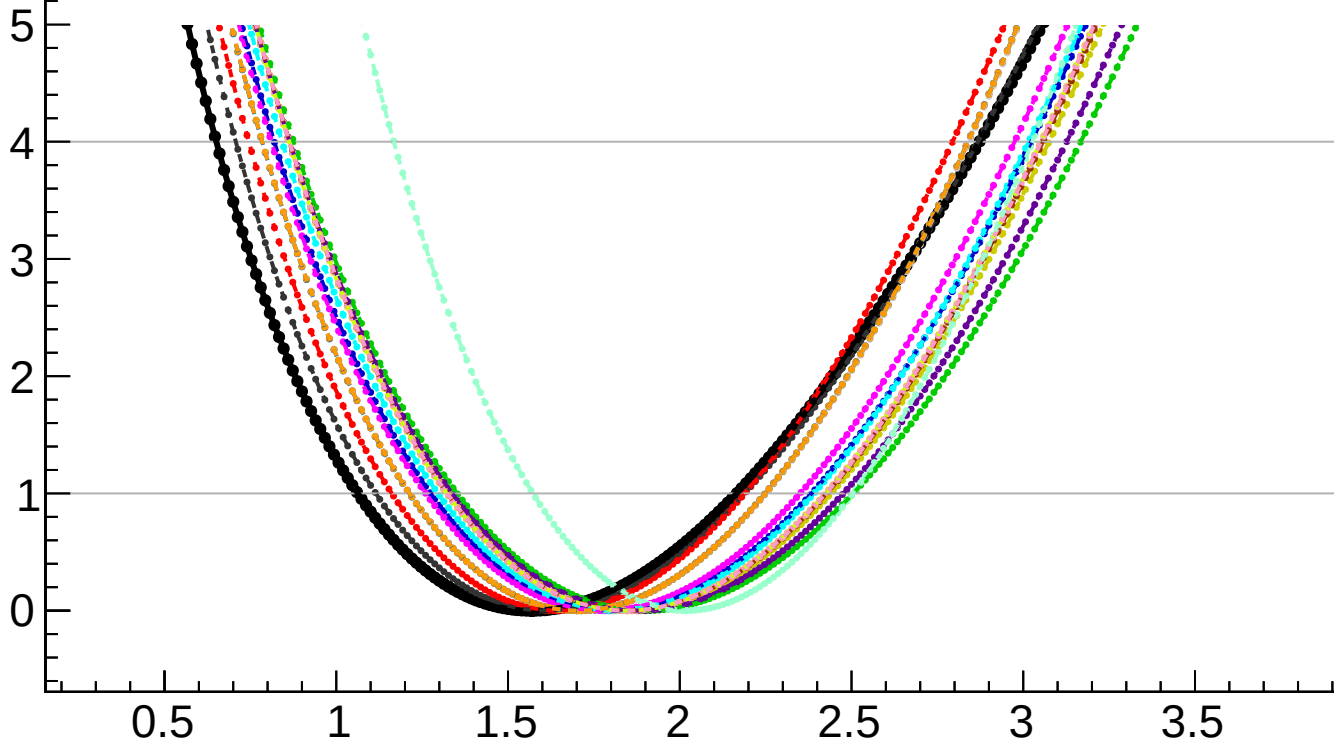
**CMS**  
*Internal*

$r = 1.569$   
 $+0.597$   
 $-0.507$

— Total uncert.  
- - - Freeze theory  
- - - Freeze OSnorm  
- - - Freeze lumi  
- - - Freeze jet  
- - - Freeze VBS  
- - - Freeze tau  
- - - Freeze mischarge

- - - Freeze FRsys  
- - - Freeze TTnorm  
- - - Freeze PF  
- - - Freeze btag  
- - - Freeze Pileup  
- - - Freeze MET  
- - - Freeze lepton  
- - - Freeze all

$r = 1.569$   $+0.144$   $-0.119$  (FRsys)  $+0.134$   $-0.112$  (theory)  $+0.030$   $-0.090$  (TTnorm)  $+0.150$   $-0.103$  (OSnorm)  $+0.117$   $-0.090$  (PF)  $+0.000$   $-0.000$  (lumi)  $+0.000$   $-0.000$  (btag)  $+0.000$   $-0.126$  (jet)  $+0.149$   $-0.126$  (Pileup)  $+0.041$   $-0.031$  (VBS)  $+0.046$   $-0.000$  (MET)  $+0.139$   $-0.105$  (tau)  $+0.023$   $-0.023$  (lepton)  $+0.000$   $-0.006$  (mischarge)  $+0.244$   $-0.209$  (MCstat)  $+0.487$   $-0.444$  (Stat)



$r$